Photonic resistive switching: Exercises

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

Unless otherwise stated, values are hypothetical, not measured device specifications. State the physical model before calculating.

E01 Modal attenuation and optical contrast

Assume a single forward mode with fixed coupling and no reflection or mode conversion. For $E \propto e^{i\beta z - i\omega t}$ and $n_{\text{eff}} = n + i\kappa$, derive $T = P(L)/P(0)$. Let $L = 2\,\mu\text{m}$, $\lambda_0 = 1.55\,\mu\text{m}$, and $\kappa_H = 0.01$, $\kappa_L = 0.08$. Calculate both transmissions, $ER = 10\log_{10}(T_H/T_L)$ and $IL_H = -10\log_{10} T_H$. Why is bulk-material κ not automatically appropriate?

E02 Pulse energy

An ideal rectangular write pulse at the device has $V = 1.2\,\text{V}$, $I = 10\,\mu\text{A}$ for $20\,\text{ns}$. Integrate its energy. Name two contributions not included in this device-only result.

E03 Channels and non-uniqueness

Use $G_0 = 77.5\,\mu\text{S}$ and the ideal coherent small-bias model $G = G_0 \sum T_n$. Calculate G and R for one channel with $T = 0.3$. Give a distinct two-channel set with the same G and explain why the observation does not uniquely reveal an atomic structure.

E04 Zero failures

For 100 independent and identically distributed observations at a fixed retention time, temperature and read protocol, no failure occurs. Derive a one-sided 95% upper bound for per-observation failure probability. Would correlated failures satisfy this model? Does it establish zero failure at all future times?

E05 Electrical and optical states

One waveguide has lower optical transmission in LRS; another MIM device has higher transmission in LRS. Explain how this is possible, and give two measurement conditions that must be fixed when comparing states.

Lin Cing https://github.com/xinhi6716/resistive-memory-self-study

Photonic resistive switching: Exercises (continued)

E06 Reflectance and dB

Reflectance changes from $R_1 = 0.20$ to $R_2 = 0.25$. Calculate the absolute difference in percentage points and $10 \log_{10}(R_2/R_1)$. Why can a five-percentage-point difference alone not determine the dB ratio?

E07 Confinement and loss

Compare qualitatively a hybrid plasmonic waveguide and a metal–insulator–metal gap. Explain how their gap fields can interact with an active region, and why greater metal overlap can increase both interaction and attenuation. State a limitation of an ideal planar-interface SPP formula.

E08 Retention protocol

Propose a write–read–wait–read sequence to test whether an optical state persists at zero applied voltage. What must be controlled and why does a voltage-sweep hysteresis loop not settle the question?

E09 Cascaded stages

Use the high-transmission state from E01 and assume three identical stages with fixed coupling and no reflection or interference. Calculate total transmission and insertion loss. Give one reason a real cascade might differ.

E10 Power, rate and energy

A system's average power over a complete observation period including active and idle intervals is $P_{\mathrm{avg}} = 12.5\,\mathrm{nW}$; the average operation rate is one per period at $f = 1\,\mathrm{MHz}$. Calculate P_{avg}/f and state whether instantaneous power must be constant. Why are peak or active-only power insufficient substitutes? Describe a measurement that distinguishes this average from a single device write-pulse energy.

E11 ZnO reflectance and evidence

For a hypothetical ZnO grating measured with fixed wavelength, polarization and reference, the low-reflection and high-reflection states give $R_1 = 0.30$ and $R_2 = 0.35$. Calculate their difference in percentage points and $10 \log_{10}(R_2/R_1)$. A TEM image shows local contrast and an EDX line scan indicates a local Zn–O composition difference. Explain what these observations support, why they do not uniquely prove an oxygen-vacancy filament, and give one complementary control or measurement.